



IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to RVK AGENCY

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

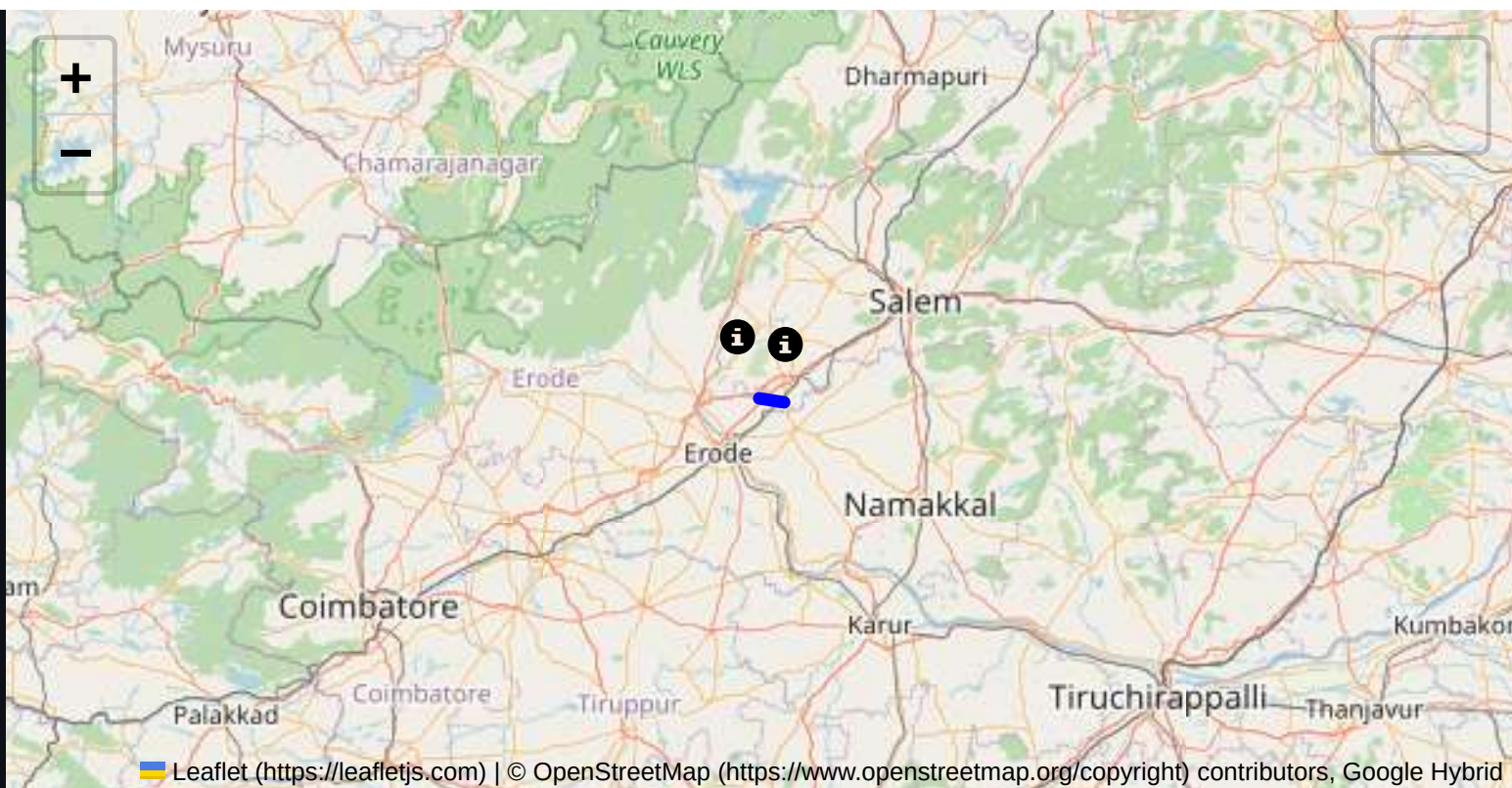
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

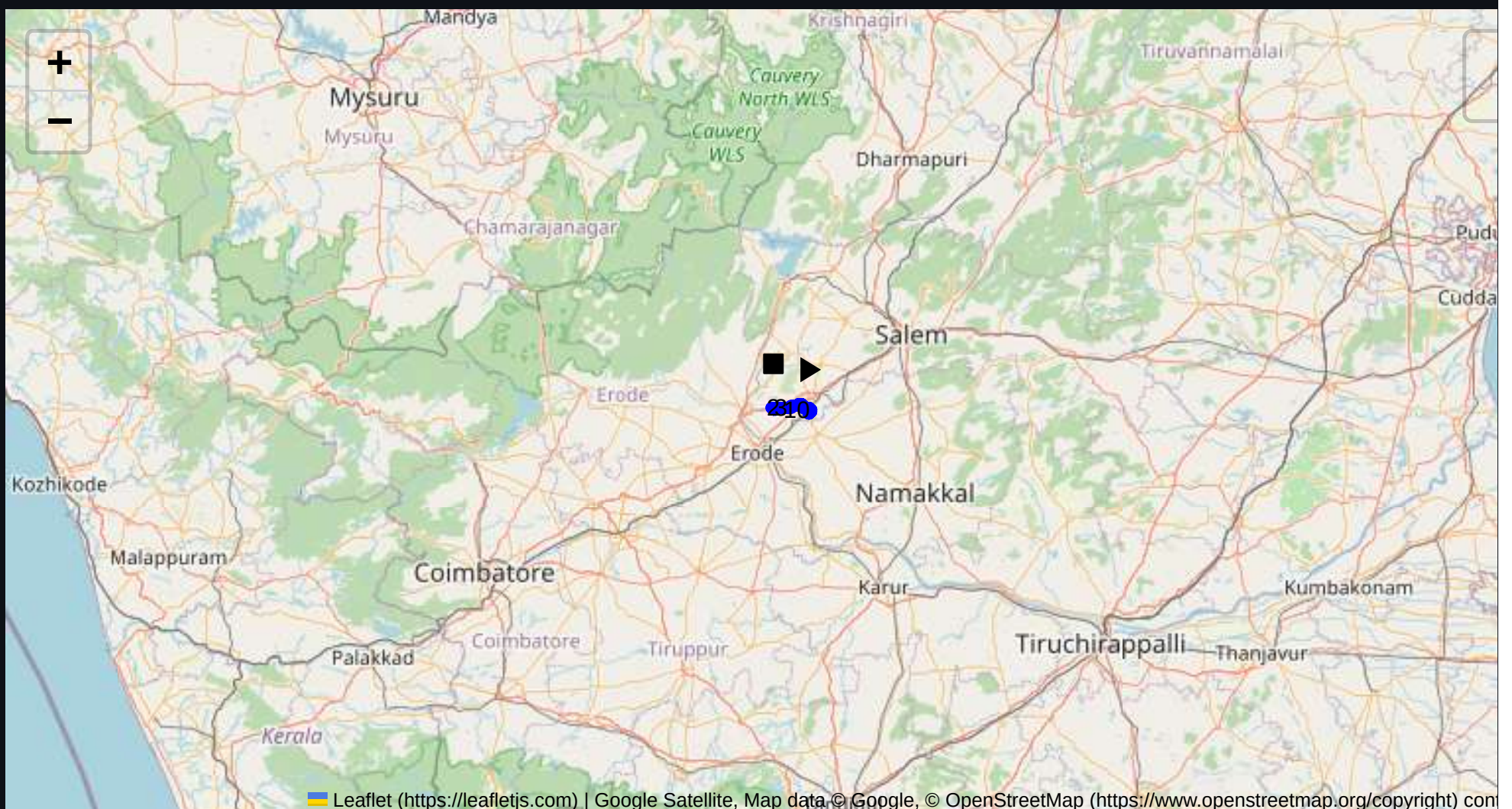
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 13.96 km**  
**Estimated Duration: 0.3 hours**  
**Adjusted Duration (Heavy Vehicle): 0.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.451373, 77.768418)**



## Welcome to the Journey Risk Management Study

### Route Safety Analysis Report

**1. Overview of the Route Map:** The route from CVQF+23W, Sangagiri to FQ29+G85, Service Rd spans approximately 13.96 kilometers within Tamil Nadu, India. This route primarily follows local roads and possibly intersects with regional highways. It is relatively short and serves as a connector between these two points.

**2. Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu experiences a tropical climate. The route could face heavy rain during monsoon seasons (June to September, and a northeast monsoon in October to December), potentially causing flooding and reduced visibility. Summers are hot and dry, with potential for heat waves affecting vehicle performance and driver comfort.

**3. Analysis of Traffic Patterns:** The route likely encounters peak traffic during morning (7:00 - 9:00 AM) and evening (5:00 - 7:00 PM) hours as it connects populated areas. Congestion may happen near towns, intersections, and any junctions with national highways. Route segments within Sangagiri and approaching Erode might occasionally experience backlogs.

**4. Assessment of Road Quality and Infrastructure:** The quality of rural and urban roads in the area can vary significantly. Some roads may have potholes or be inadequately maintained, especially post-monsoon. Key infrastructure such as bridges and traffic signals should be closely monitored for operational status. The route may also include narrow lanes that are challenging for heavy vehicles.

**5. Suggestions for Alternative Routes for Emergencies:** If the primary route is blocked or becomes unsafe, an alternate route via alternative local roads or highways should be predetermined, depending on live updates. Maps and GPS should be utilized but must be confirmed for truck accessibility and hazardous material restrictions.

**6. Summary of Local Regulations Affecting Hazardous Material Transport:** Transporting hazardous materials in Tamil Nadu requires adherence to legal guidelines involving route permits, vehicle permits, and time restrictions. Night-time travel might be restricted for hazardous materials to minimize risk to other road users.

**7. Overview of Historical Incidents:** Historically, incidents involving heavy vehicles on similar routes in Tamil Nadu have included accidents due to over-speeding, poor road conditions, and insufficient signaling during adverse weather. Rollovers and spills have also occurred where gradient and road condition factored negatively.

**8. Environmental Considerations and Sensitive Areas:** There might be environmentally sensitive areas along the route, such as agricultural lands, water bodies, and settlements that must be protected from potential leaks or spills. Proper containment and emergency response plans are crucial for such stretches.

**9. Analysis of Communication Coverage:** While mobile coverage is generally reliable in urban areas along the route, rural segments might experience occasional dead zones. It is advisable to carry alternative communication tools or pre-download maps for navigation assistance.

**10. Estimated Emergency Response Times:** Emergency response times vary depending on location, with more urban segments likely to receive quicker assistance compared to rural stretches. The response in rural areas could take 30 minutes to over an hour, while urban areas may see times closer to 15-30 minutes.

**11. Overall Summary of Risk Assessment:** The route itself presents moderate risk due to traffic volumes, road conditions, and weather patterns typical for the region. Key hazards include potential for congestion, lack of infrastructure in more remote segments, and weather-related challenges during monsoon seasons. Ensuring that drivers have up-to-date information, adhere to safety regulations, and have contingency plans for communication and rerouting will mitigate risks significantly. Consideration for environmental

impact and emergency readiness remains paramount for heavy vehicle drivers transporting hazardous materials.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 1 | Turn       | Medium     | 11.43956, 77.87340 | 30 KM/Hr    |
| 2 | Turn       | Medium     | 11.43971, 77.87348 | 30 KM/Hr    |
| 3 | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4 | Turn       | Medium     | 11.46307, 77.84941 | 30 KM/Hr    |
| 5 | Turn       | Medium     | 11.46318, 77.84913 | 30 KM/Hr    |
| 6 | Turn       | High       | 11.45517, 77.81345 | 15 KM/Hr    |
| 7 | Turn       | High       | 11.45521, 77.81343 | 15 KM/Hr    |
| 8 | Turn       | High       | 11.45085, 77.77020 | 15 KM/Hr    |
| 9 | Blind Spot | Blind Spot | 11.45111, 77.77024 | 10 KM/Hr    |

### Emergency Locations

|   | type     | name                                 | coordinates            | speed_limit | risk_level |
|---|----------|--------------------------------------|------------------------|-------------|------------|
| 3 | hospital | Government Hospital                  | 11.4525596, 77.7749426 | 30 km/h     | Medium     |
| 4 | hospital | Dhanvantri Multi Speciality Hospital | 11.451362, 77.766602   | 30 km/h     | Medium     |

### Crowded Spots

|   | type   | name                       | coordinates            | speed_limit | risk_level |
|---|--------|----------------------------|------------------------|-------------|------------|
| 0 | school | KRP Matric. Hr. Sec School | 11.4546193, 77.8142445 | 30 km/h     | Medium     |

|   | type        | name                             | coordinates               | speed_limit | risk_level |
|---|-------------|----------------------------------|---------------------------|-------------|------------|
| 1 | college     | Vivekanandha Engineering College | 11.4589312,<br>77.7899284 | 30 km/h     | Medium     |
| 2 | marketplace | Monday market                    | 11.452863, 77.775989      | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46307, 77.84941



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46318, 77.84913



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45517, 77.81345



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45521, 77.81343



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45085, 77.77020



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.45111, 77.77024

